

HIGH-PERFORMANCE COMPUTING ON A BEAGLE BOARD

The Beagle Board is a generic board powered by an OMAP3530 processor. It is backed by a highly active community, which maintains its own portal that invites discussions, queries, ideas and other inputs from all over the world—the first of its kind

■ ATANU DATTA AND ABHIJIT PAUL CHOUDHURY

The Beagle Board—an open source initiative by Texas Instruments (TI)—was launched to the public around a year ago. It was developed by a small team at Texas Instruments, led by Khasim Syed Mohammed, who's the lead developer for open platforms at TI. It's called the

Beagle Board because of its dimensions. A beagle is a breed of short and cute dogs, which the team found similar to this 7.6×7.6cm² (3×3-inch²) board. It's a low-cost, fan-less, single-board computer with a high-performance embedded platform that is based on Texas Instruments' OMAP3530 technology.

"The idea of a Beagle Board was born a year back and I have been involved in this initiative from the very beginning. It took six months for the team to convert this idea into a reality. I started with hardware enablement for this initiative with boot loaders, the Linux kernel and drivers, validation, documentation, promotions, etc," notes Syed. "Ever since its launch, I've been promoting Beagle Board among university students and the small-scale IT sector in India, continuing in enabling a global community to be built around OMAP3 technology. Apart from other activities, I've also been working to demonstrate the capability of OMAP3 architecture in the medical, infotainment, portable navigation and low-power industrial sectors."

The interesting point to note here

is that the board is targeted at the open source developers, which seemed obvious considering the lead developer's passion for open source. "I was interested in computers right from my early days. After my school, I started learning computers and during my internship on TI's DSP, my curiosity for embedded applications increased enormously. Moving forward, I found open source to be more useful since it was free and gave me all the necessary software blocks 'with source' to build end-to-end solutions," reveals Syed. "I started as a user/consumer of open source software and forums. While working closely with the OMAP Linux community, I found significant gaps in our offerings to the open community. I joined Jason Kridner and Gerald Coley at TI to bridge and address these gaps through beagleboard.org."

He adds: "I am an enabler and a contributor of open software in the community. The demand for consumer devices is growing at a fast rate and we need to match this pace of demand and supply. Open source initiatives will be playing a major role in building an open ecosystem to enable innovations, implementing them in products and to meet the global demand for consumer products. These are some of the areas that I've been trying to cater to through the Beagle Board initiative."

Open source software is, in fact, embedded into the board—it consists

